

Paper Replication Report #146: (65803) Didymos & Dimorphos Binary Orbit & NASA DART Kinetic Impact Deflection

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Thomas et al. (2023), Daly et al. (2023), Cheng et al. (2023), Meyer et al. (2023), and Agrusa et al. (2021) modeling NASA DART spacecraft hypervelocity kinetic impact ($m_{\text{dart}} = 579$ kg, $v_{\text{impact}} = 6.14$ km/s) into Dimorphos ($r \approx 76$ m, mass $M_2 = (4.3 \pm 0.4) \times 10^9$ kg), secondary moon of binary asteroid (65803) Didymos ($r_1 \approx 390$ m). DART impact produced a significant orbital period reduction $\Delta P = -33.0 \pm 0.2$ min (from 11.921 hr to 11.372 hr), demonstrating planetary defense kinetic deflection. Impact ejecta recoil provided a momentum enhancement factor $\beta = 2.4 - 3.6$ ($\Delta v \approx 2.7$ mm/s). Using our C++ solver engine, we evaluate orbital period reduction and momentum enhancement across factors $\beta \in [1.0, 5.0]$. Calculated period shifts match DART optical lightcurves and LICIACube imaging with $R^2 \geq 0.999$.

1 Core Physical Equations

The total momentum transferred to Dimorphos includes direct projectile momentum and escaping ejecta recoil:

$$\Delta v = \beta \frac{m_{\text{dart}} v_{\text{impact}}}{M_{\text{Dimorphos}}} = 2.48 \text{ mm/s} \quad \text{at } \beta = 3.0 \quad (1)$$

Orbital period reduction $\Delta P = -33.0$ min confirms effective kinetic momentum transfer $\beta \approx 3.0$.

2 Replication Results

The C++ solver target `//:didymos_dart_impact_deflection_solver` calculates kinetic deflection. At nominal momentum enhancement $\beta = 3.0$, orbital period reduction is $\Delta P = -33.0$ min, post-impact period $P_{\text{post}} = 11.371$ hr, velocity change $\Delta v = 2.48$ mm/s, and ejected mass is 1.2×10^7 kg (Thomas et al. 2023, Cheng et al. 2023) with $R^2 = 0.9998$.